

Firmware Deployment Checklist

Document Control

Document Information	Details
Document ID	[Unique ID - e.g., FDC-2025-001]
Related Firmware Update ID	[Reference to Firmware Update Planning document]
Version	[e.g., 1.0]
Created By	[Name and Role]
Creation Date	[YYYY-MM-DD]

Deployment Information

Item	Details
Device Type	[e.g., ASIC Miner, Network Switch, PDU, etc.]
Make and Model	[e.g., Bitmain Antminer S19 Pro]
Current Firmware Version	[e.g., v2.0.1.3]
Target Firmware Version	[e.g., v2.0.1.5]
Deployment Date	[YYYY-MM-DD]
Deployment Time Window	[Start Time - End Time]
Deployment Type	☐ Standard ☐ Emergency ☐ Critical
Deployment Method	☐ Phased ☐ Canary ☐ Batch ☐ All-at-once
Affected Devices/Count	[e.g., 250 units in Racks A1-D10]

Deployment Team

Role	Name	Contact Information	Responsibilities
Deployment Lead			
Technical Support			
Operations Monitor			
Security Observer			
Escalation Contact			

1. Pre-Deployment Verification

1.1 Planning and Approvals

Task	Status	Completed By	Date	Notes
Firmware Update Plan approved	☐ Complete ☐ N/A			
Testing documentation reviewed and approved	☐ Complete ☐ N/A			
All required approvals obtained	☐ Complete ☐ N/A			
Deployment schedule communicated to stakeholders	☐ Complete ☐ N/A			
Maintenance window confirmed and scheduled	☐ Complete ☐ N/A			
Mining pool notifications sent (if applicable)	☐ Complete ☐ N/A			

1.2 Firmware Verification

Task	Status	Completed By	Date	Notes
Firmware file obtained from authorized source	☐ Complete ☐ N/A			
Firmware file checksum/hash verified	☐ Complete ☐ N/A			
Digital signature verified (if applicable)	☐ Complete ☐ N/A			
Firmware compatibility verified	☐ Complete ☐ N/A			
Release notes and known issues reviewed	☐ Complete ☐ N/A			

1.3 Backup Verification

Task	Status	Completed By	Date	Notes
Current firmware version backed up	☐ Complete ☐ N/A			
Device configurations backed up	☐ Complete ☐ N/A			
Backup copies verified and accessible	☐ Complete ☐ N/A			
Backup stored in secure location	☐ Complete ☐ N/A			
Rollback procedure confirmed and tested	☐ Complete ☐ N/A			

1.4 Environment and Resource Verification

Task	Status	Completed By	Date	Notes
Network connectivity confirmed	☐ Complete ☐ N/A			
Required bandwidth available	☐ Complete ☐ N/A			
Power stability verified	☐ Complete ☐ N/A			
All deployment tools available and tested	☐ Complete ☐ N/A			
Deployment team members available and briefed	☐ Complete ☐ N/A			
Monitoring systems operational	☐ Complete ☐ N/A			

1.5 Device Status Verification

Task	Status	Completed By	Date	Notes
Current device status documented	☐ Complete ☐ N/A			
Performance baseline established	☐ Complete ☐ N/A			
Devices operating normally	☐ Complete ☐ N/A			
No pending issues with target devices	☐ Complete ☐ N/A			
Devices accessible for update	☐ Complete ☐ N/A			

2. Deployment Execution

2.1 Pre-Deployment Final Check

Task	Status	Completed By	Date	Notes
Go/No-Go decision made	☐ Complete ☐ N/A			
All team members in position	☐ Complete ☐ N/A			
Communication channels established	☐ Complete ☐ N/A			
Monitoring systems confirmed operational	☐ Complete ☐ N/A			
Deployment steps reviewed with team	☐ Complete ☐ N/A			
Rollback criteria and procedure confirmed	☐ Complete ☐ N/A			

2.2 Initial Deployment (Canary/First Batch)

Task	Status	Completed By	Date	Notes
Initial devices selected	☐ Complete ☐ N/A			
Initial deployment executed	☐ Complete ☐ N/A			
Initial verification performed	☐ Complete ☐ N/A			
No critical issues observed	☐ Complete ☐ N/A			
Decision to proceed with full deployment	☐ Complete ☐ N/A			

2.3 Full Deployment

Task	Status	Completed By	Date	Notes
Full deployment commenced	☐ Complete ☐ N/A			
Deployment progress tracked	☐ Complete ☐ N/A			
Deployment milestones achieved	☐ Complete ☐ N/A			
All devices successfully updated	☐ Complete ☐ N/A			
Any failures or exceptions documented	☐ Complete ☐ N/A			

2.4 Deployment Monitoring

Task	Status	Completed By	Date	Notes
Performance monitored during deployment	☐ Complete ☐ N/A			
Error logs monitored	☐ Complete ☐ N/A			
Network stability monitored	☐ Complete ☐ N/A			
Hashrate/production impact tracked	☐ Complete ☐ N/A			
Any anomalies documented	☐ Complete ☐ N/A			

3. Post-Deployment Verification

3.1 Functionality Verification

Task	Status	Completed By	Date	Notes
All devices operating properly	☐ Complete ☐ N/A			
New firmware version confirmed on all devices	☐ Complete ☐ N/A			
All functions operating as expected	☐ Complete ☐ N/A			
No unexpected errors or warnings	☐ Complete ☐ N/A			
User interfaces functioning correctly	☐ Complete ☐ N/A			

3.2 Performance Verification

Task	Status	Completed By	Date	Notes
Hashrate verified against baseline	☐ Complete ☐ N/A			
Power consumption verified	☐ Complete ☐ N/A			
Temperature levels verified	☐ Complete ☐ N/A			
Efficiency metrics calculated	☐ Complete ☐ N/A			
Network performance verified	☐ Complete ☐ N/A			
Application/service performance verified	☐ Complete ☐ N/A			

3.3 Integration Verification

Task	Status	Completed By	Date	Notes
Network connectivity verified	☐ Complete ☐ N/A			
Mining pool connectivity verified	☐ Complete ☐ N/A			
Management systems integration verified	☐ Complete ☐ N/A			
Monitoring systems receiving data	☐ Complete ☐ N/A			
Alert systems functioning properly	☐ Complete ☐ N/A			
All dependent systems operating normally	☐ Complete ☐ N/A			

3.4 Security Verification

Task	Status	Completed By	Date	Notes
Access controls verified	☐ Complete ☐ N/A			
Authentication mechanisms tested	☐ Complete ☐ N/A			
Security settings verified	☐ Complete ☐ N/A			
No new vulnerabilities detected	☐ Complete ☐ N/A			
Security scans performed (if applicable)	☐ Complete ☐ N/A			

4. Issue Management and Rollback

4.1 Issue Tracking

Issue ID	Description	Severity	Status	Resolution	Notes
[Add more as needed]					

4.2 Rollback Execution (if needed)

Task	Status	Completed By	Date	Notes
Rollback decision made	☐ Complete ☐ N/A			
Rollback procedure initiated	☐ Complete ☐ N/A			
Previous firmware restored	☐ Complete ☐ N/A			
Configuration restored	☐ Complete ☐ N/A			
Functionality verified after rollback	☐ Complete ☐ N/A			
Performance verified after rollback	☐ Complete ☐ N/A			

5. Deployment Completion

5.1 Final Documentation

Task	Status	Completed By	Date	Notes
Deployment results documented	☐ Complete ☐ N/A			
Inventory systems updated	☐ Complete ☐ N/A			
Issues documented in knowledge base	☐ Complete ☐ N/A			
Performance changes documented	☐ Complete ☐ N/A			
Lessons learned documented	☐ Complete ☐ N/A			

5.2 Stakeholder Communication

Task	Status	Completed By	Date	Notes
Deployment completion notification sent	☐ Complete ☐ N/A			
Performance impact communicated	☐ Complete ☐ N/A			
Any ongoing issues communicated	☐ Complete ☐ N/A			
Follow-up actions communicated	☐ Complete ☐ N/A			
Management summary prepared	☐ Complete ☐ N/A			

5.3 Deployment Sign-off

Role	Name	Signature	Date
Deployment Lead			
Technical Lead			
Operations Manager			
Security Officer			
IT/Infrastructure Manager			

6. Extended Monitoring Period

6.1 Extended Monitoring Plan

Monitoring Period	Start Date	End Date	Key Metrics to Monitor	Responsible Person
24 Hours				
72 Hours				
1				

6.1 Extended Monitoring Plan

Monitoring Period	Start Date	End Date	Key Metrics to Monitor	Responsible Person
24 Hours				
72 Hours				
1 Week				
2 Weeks				
1 Month				

6.2 Monitoring Checkpoints

Checkpoint	Date	Status	Findings	Action Required
24-Hour Check		☐ Normal ☐ Issues Found		
72-Hour Check		☐ Normal ☐ Issues Found		
1-Week Check		☐ Normal ☐ Issues Found		
2-Week Check		☐ Normal ☐ Issues Found		
1-Month Check		☐ Normal ☐ Issues Found		

6.3 Extended Monitoring Completion

Task	Status	Completed By	Date	Notes
Extended monitoring completed	☐ Complete ☐ N/A			
No significant issues identified	☐ Complete ☐ N/A			
Performance stable throughout period	☐ Complete ☐ N/A			
Final report prepared	☐ Complete ☐ N/A			
Deployment fully closed	☐ Complete ☐ N/A			

7. Deployment Metrics Summary

7.1 Time Metrics

Metric	Planned	Actual	Variance	Notes
Preparation Time				
Deployment Duration				
Verification Time				
Total Downtime				
Recovery Time (if needed)				

7.2 Quality Metrics

Metric	Target	Actual	Within Target?	Notes
Devices Successfully Updated	100%		☐ Yes ☐ No	
First-Attempt Success Rate	>95%		☐ Yes ☐ No	
Issues Requiring Intervention	<5%		☐ Yes ☐ No	
Rollbacks Required	0		☐ Yes ☐ No	
Post-Deployment Issues	<2%		☐ Yes ☐ No	

7.3 Performance Impact

Metric	Pre-Update	Post-Update	Change	Impact
Average Hashrate				☐ Positive ☐ Neutral ☐ Negative
Power Consumption				☐ Positive ☐ Neutral ☐ Negative
Efficiency (J/TH)				☐ Positive ☐ Neutral ☐ Negative

Operating Temperature				☐ Positive ☐ Neutral ☐ Negative
Rejection Rate				☐ Positive ☐ Neutral ☐ Negative

8. Lessons Learned

8.1 What Went Well

[Document aspects of the deployment that were successful]

8.2 What Could Be Improved

[Document areas where improvements could be made in future deployments]

8.3 Specific Improvement Actions

Improvement Area	Action Item	Owner	Target Date
[Add more as needed]			

Attachment Checklist

- ☐ Deployment Logs
- ☐ Performance Data (Before and After)
- ☐ Issue Documentation
- ☐ Screenshots/Evidence
- ☐ Team Communications
- ☐ Management Reports
- ☐ Other: _____

Note: This checklist should be completed for all firmware deployments. For emergency deployments, some sections may be abbreviated but must still be addressed post-deployment. Retain this documentation according to the organization's record retention policy.